

Total No. of Questions : 4]

SEAT No. :

PF297

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Apr-26/TE/Insem-377

T.E. (Computer Engineering) (Insem)
DATA SCIENCE AND BIG DATA ANALYTICS
(2019 Pattern) (Semester - VI) (310251)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2 and Q3 or Q4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

- Q1)** a) What is the need of Data Science and Big data analytics? Explain any three sources of big data. [5]
- b) Suppose that the data for analysis includes the attribute age given the following data (in increasing order) for the attribute age: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. [5]
- i) Use smoothing by bin means, using a bin depth of 3.
- ii) What other methods are there for data smoothing?
- c) What is data discretization? What are the different techniques used for data discretization. [5]

OR

- Q2)** a) The following data are a list of All Electronics prices for commonly sold items (rounded to the nearest dollar). The numbers have been sorted: 1, 1, 5, 5, 5, 5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18, 18, 18, 18, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30, 30, 30.
- Show a histogram for the data using singleton buckets. How can we further reduce the data. [5]
- b) What are the challenges of Data science and Big data? What are the 3 main characteristics of Big Data. [5]
- c) Explain data science life cycle. [5]

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- Q3) a)** What is T-tests. What are its types explain. [5]
- b) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. [5]
- i) What is the mean of the data? What is the median?
- ii) What is the mode of the data? Comment on the data's modality (i.e., bimodal, trimodal, etc.).
- iii) What is the midrange of the data?
- iv) Can you find (roughly) the first quartile (Q1) and the third quartile (Q3) of the data?
- c) Describe the Chi-Square Test of Independence. [5]

OR

- Q4) a)** With an example, explain one-tailed & two-tailed t-tests. [5]
- b) What is Pearson Correlation explain. When it is used. [5]
- c) A company specializing in parachute assembly states that its main parachute failure rate is not more than 1%. You perform a hypothesis test to determine whether the company's claim is false. When will a type I or type II error occur? Which is more serious?

State the null and alternative hypotheses. [5]

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